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Started	Thursday, 29 October 2026, 11:55 PM
Completed	Friday, 30 October 2026, 12:00 AM
Duration	5 mins 23 secs
Grade	**13.00** out of 15.00 (**86.67**%)

Question **1**

Complete

Mark 1.00 out of 1.00

What is a significant characteristic of Bubble Sort?

a. It uses the heap data structure
b. It bubbles up the largest element in each pass
c. It divides the list into sublists
d. It recursively sorts subproblems

The correct answer is: It bubbles up the largest element in each pass

Question **2**

Complete

Mark 1.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

a. 40,80,90,150,200,300,400,470,70,120
b. 80,150,200,470,40,90,300,400,70,120
c. 40,70,80,90,120,150,200,300,400,470
d. 200,470,80,150,40,90,300,400,70,120

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **3**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the term "sorting" in computer science?

a. Removing duplicates from a list
b. Merging two datasets
c. Finding a specific element in a list
d. Arranging data in a specific order

The correct answer is: Arranging data in a specific order

Question **4**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after dividing the input into smaller parts?

a. Each part is searched for a specific element
b. Each part is sorted independently
c. The parts are ignored
d. The parts are merged without sorting

The correct answer is: Each part is sorted independently

Question **5**

Complete

Mark 1.00 out of 1.00

Algorithm design technique used in merge sort algorithm is

a. Greedy method
b. Divide and conquer
c. Dynamic programming
d. Backtracking

The correct answer is:
Divide and conquer

Question **6**

Complete

Mark 1.00 out of 1.00

Which Python function would you use to sort a list in-place?

- a. arrange()
- b. sorted()
- c. order()
- d. sort()

The correct answer is: sort()

Question **7**

Complete

Mark 1.00 out of 1.00

What is a key disadvantage of Bubble Sort compared to more advanced algorithms like Merge Sort?

- a. Bubble Sort is difficult to implement
- b. Bubble Sort is less efficient for large lists
- c. Bubble Sort cannot handle duplicate elements
- d. Bubble Sort does not guarantee sorted order

The correct answer is: Bubble Sort is less efficient for large lists

Question **8**

Complete

Mark 0.00 out of 1.00

Very slow way of sorting is_____

- a. Bubble sort
- b. Quick sort
- c. Heap sort
- d. Insertion sort

The correct answer is:
Insertion sort

Question **9**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm would be preferred for its divide-and-conquer approach?

a. Linear Search
b. Bubble Sort
c. Merge Sort
d. Binary Search

The correct answer is: Merge Sort

Question **10**

Complete

Mark 0.00 out of 1.00

Which algorithm is efficient for analyzing the frequency distribution of items in a list?

a. Linear Search
b. Merge Sort
c. Bubble Sort
d. Quick Sort

The correct answer is: Merge Sort

Question **11**

Complete

Mark 1.00 out of 1.00

What does the Bubble Sort algorithm primarily focus on during each pass?

a. Bubbling up the smallest element
b. Dividing the list into halves
c. Bubbling up the largest element to its correct position
d. Sorting the entire list in one pass

The correct answer is: Bubbling up the largest element to its correct position

Question **12**

Complete

Mark 1.00 out of 1.00

What is mean by stable sorting algorithm?

a. A sorting algorithm is stable if it preserves the order of all keys

b. A sorting algorithm is stable if it doesn't preserver the order of duplicate keys

c. A sorting algorithm is stable if it preserves the order of non-duplicate keys

d. A sorting algorithm is stable if it preserves the order of duplicate keys

The correct answer is:
A sorting algorithm is stable if it preserves the order of duplicate keys

Question **13**

Complete

Mark 1.00 out of 1.00

_____ explain how an algorithm will perform when the input grows larger.

a. Searching

b. Complexity

c. Sorting

d. Merging

The correct answer is:
Complexity

Question **14**

Complete

Mark 1.00 out of 1.00

Why is sorting important for selection operations?

a. It makes the data unsorted

b. It complicates the selection of items

c. It makes it easier to select items based on their relationship to the rest of the items

d. It slows down the process

The correct answer is: It makes it easier to select items based on their relationship to the rest of the items

Question **15**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the process of Merge Sort?

a. It compares adjacent elements and swaps them if necessary
b. It builds a sorted array one element at a time
c. It divides the list into two halves, sorts each half, and then merges them
d. It repeatedly finds the minimum element and moves it to the sorted part of the list

The correct answer is: It divides the list into two halves, sorts each half, and then merges them